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(54) **COMPOSITE MATERIAL, FLYING BODY
AND COMPOSITE MATERIAL
MANUFACTURING METHOD**

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(2019.01); **B32B 18/00** (2013.01);
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(58) **Field of Classification Search**

CPC C04B 35/80

(Continued)

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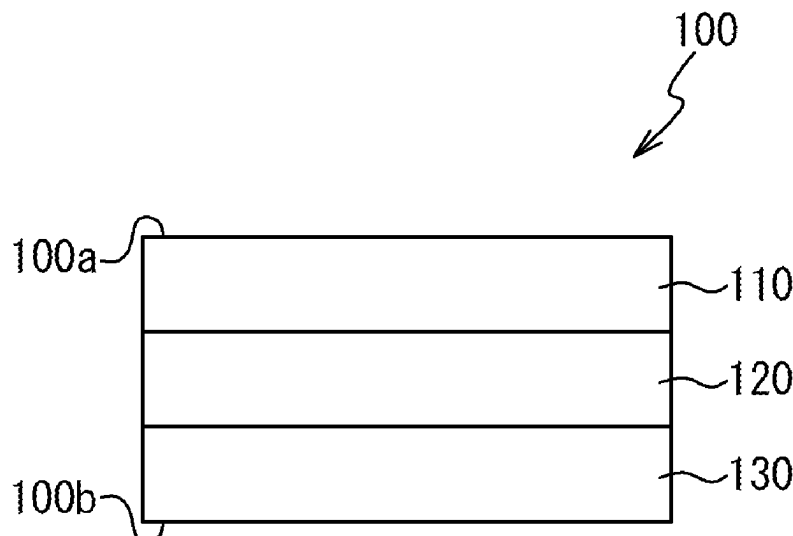
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(57) **ABSTRACT**

A composite material manufacturing method includes: lami-
nating a first sheet (210) including a first slurry (214) and a
third sheet (230) including a third slurry (234); and heating
the first sheet (210) and the third sheet (230) that are
laminated to a temperature of transforming to ceramics by
pyrolysis to form an intermediate body (300). The manu-
facturing method further includes impregnating the interme-
diate body (300) with a slurry and heating at a temperature
lower than a temperature of transforming to ceramics by
pyrolysis.

15 Claims, 8 Drawing Sheets



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C04B 35/628 (2006.01)
C04B 35/64 (2006.01)
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CPC **C04B 35/62844** (2013.01); **C04B 35/64** (2013.01); **C04B 2235/5256** (2013.01); **C04B 2235/616** (2013.01); **C04B 2235/661** (2013.01) JP 2011-522099 7/2011
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FIG. 1

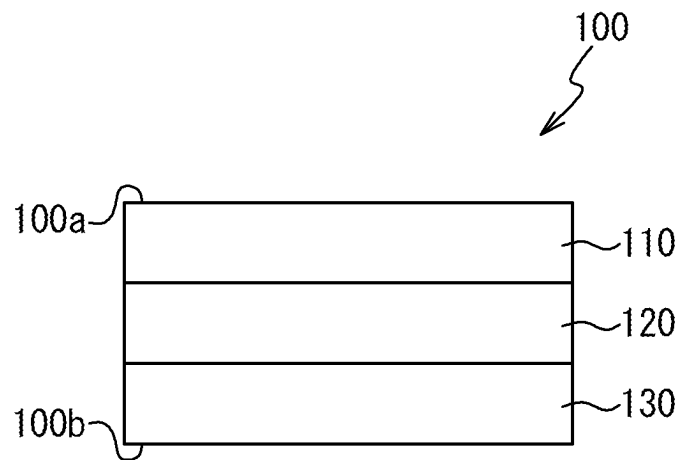


FIG. 2A

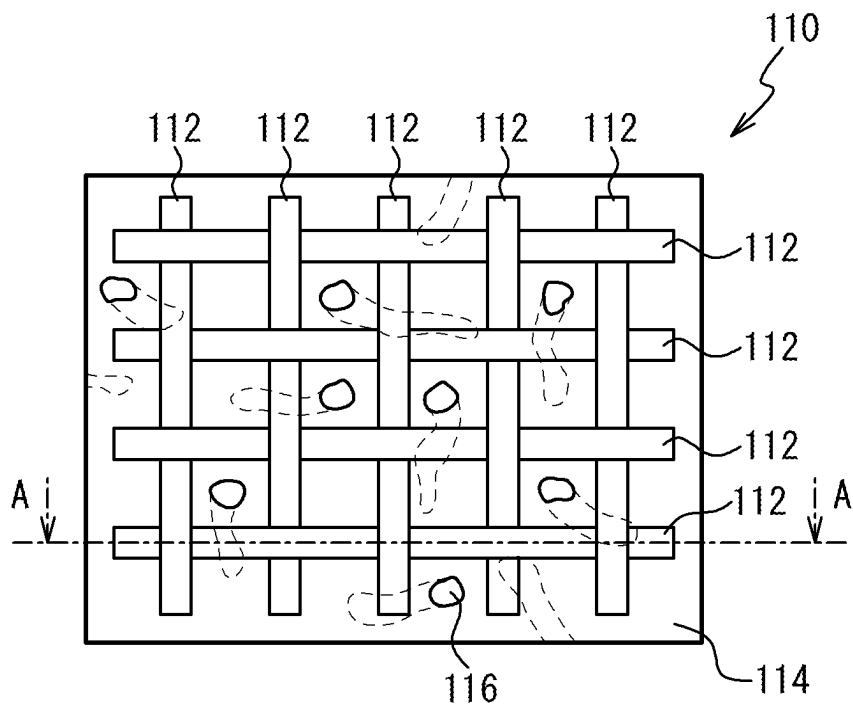


FIG. 2B

A-A Cross Section

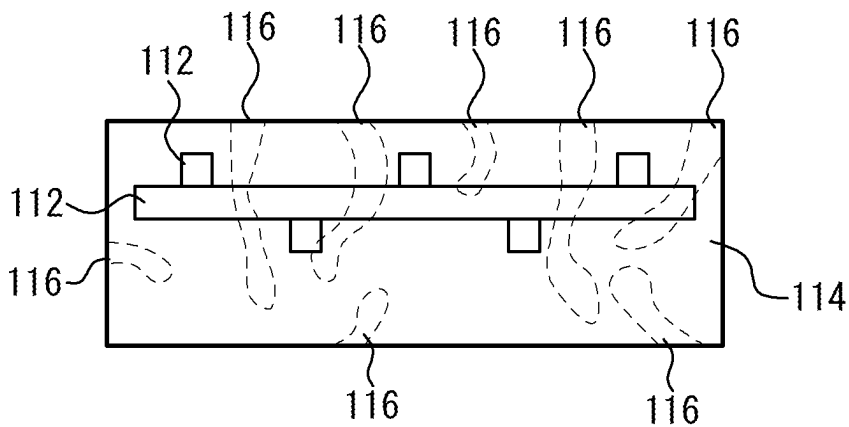


FIG. 3

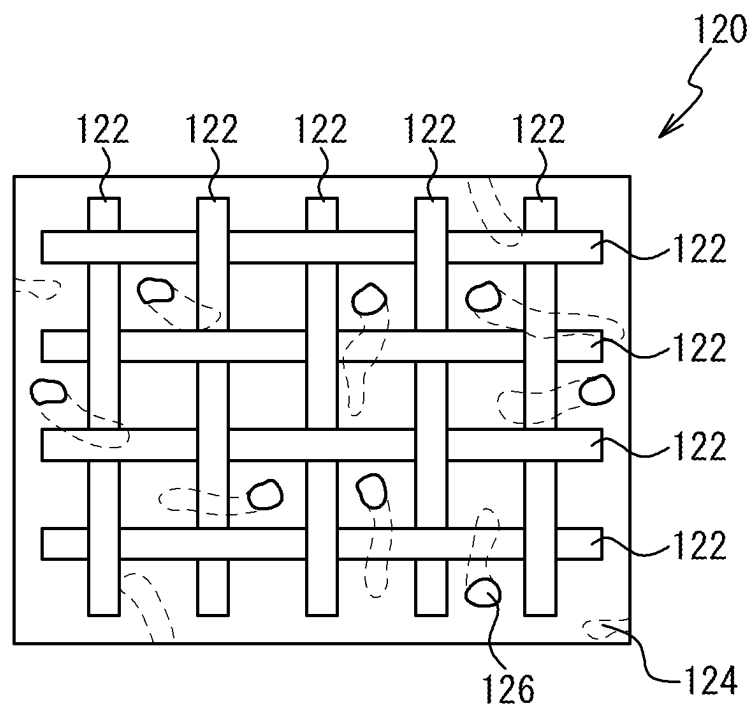


FIG. 4

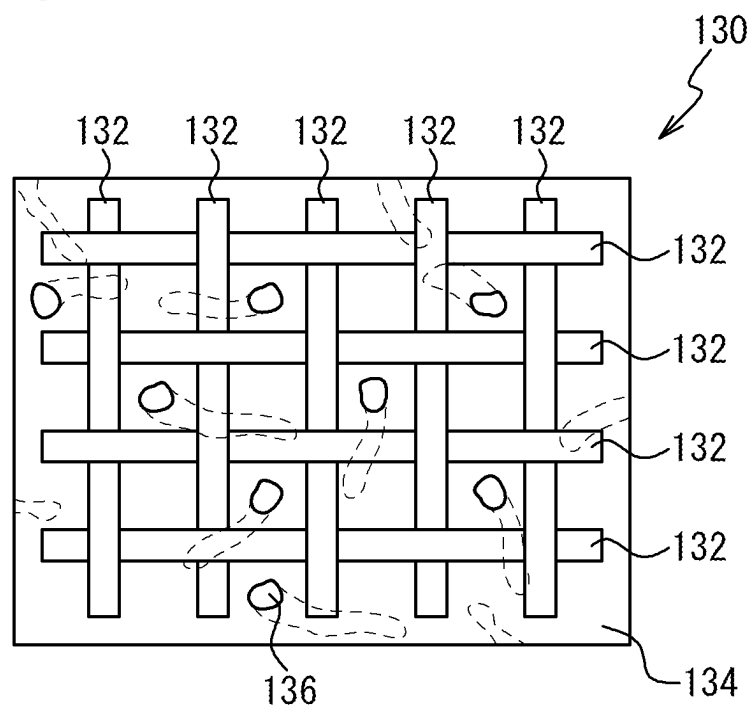


FIG. 5

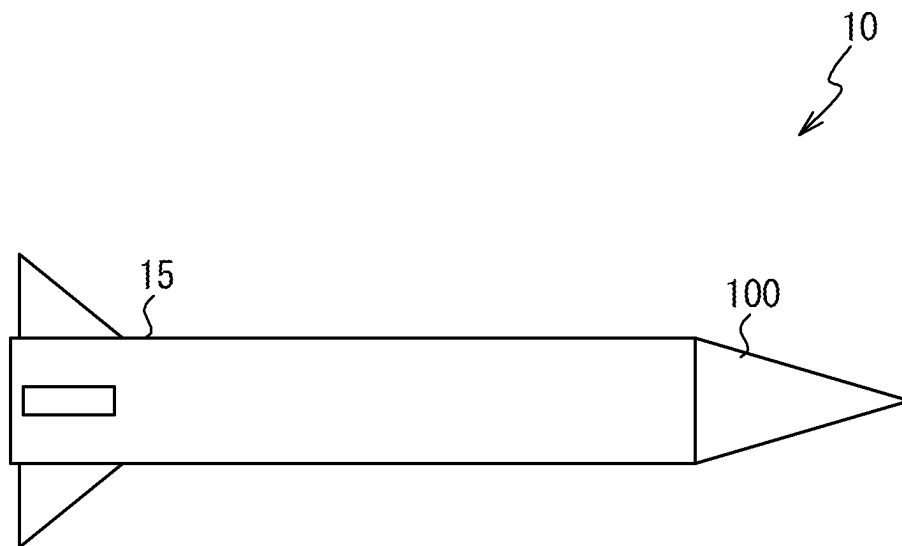


FIG. 6

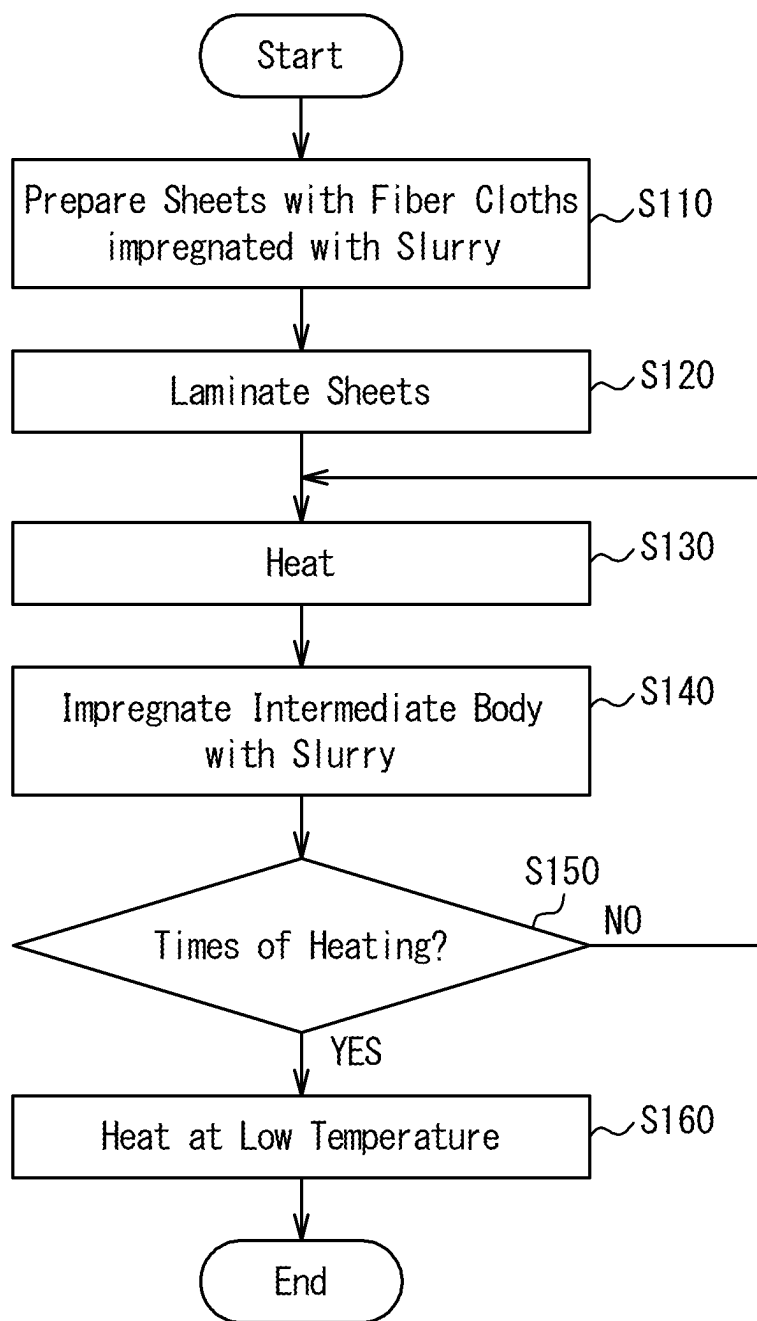


FIG. 7

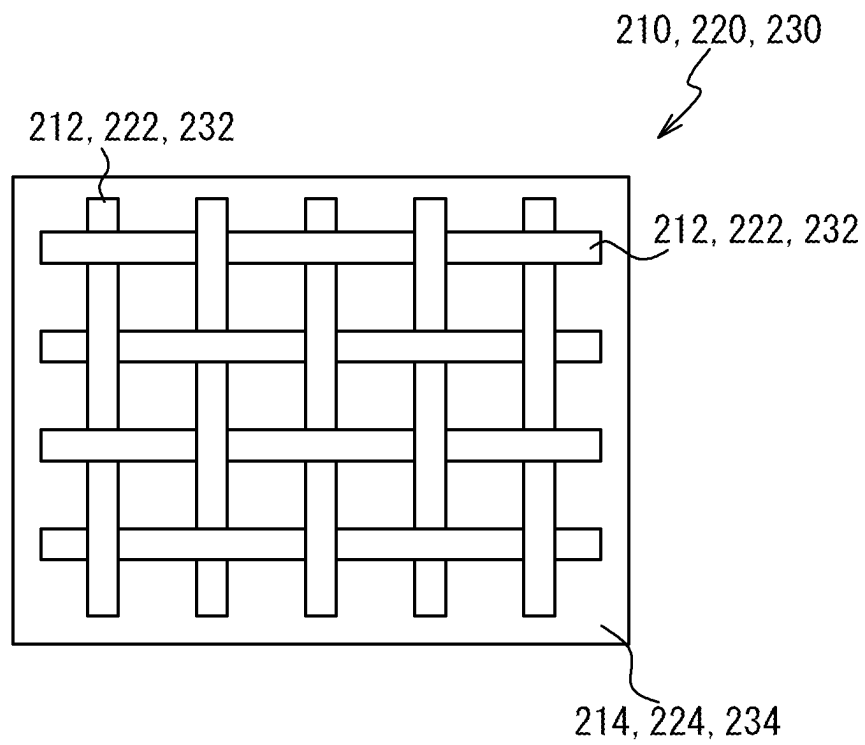


FIG. 8

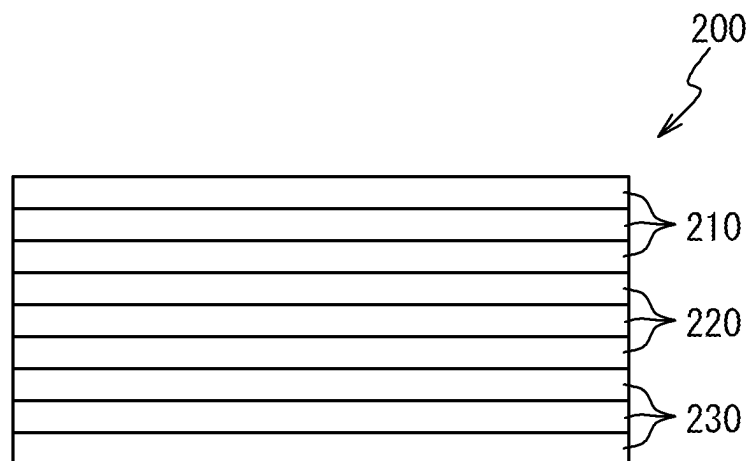


FIG. 9

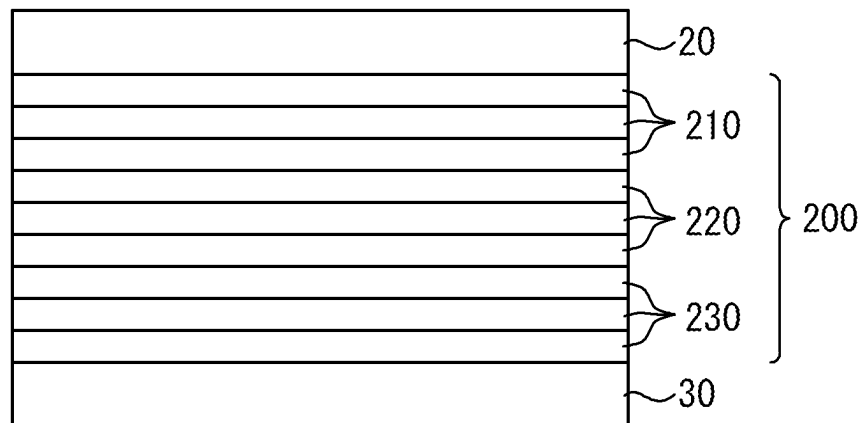


FIG. 10

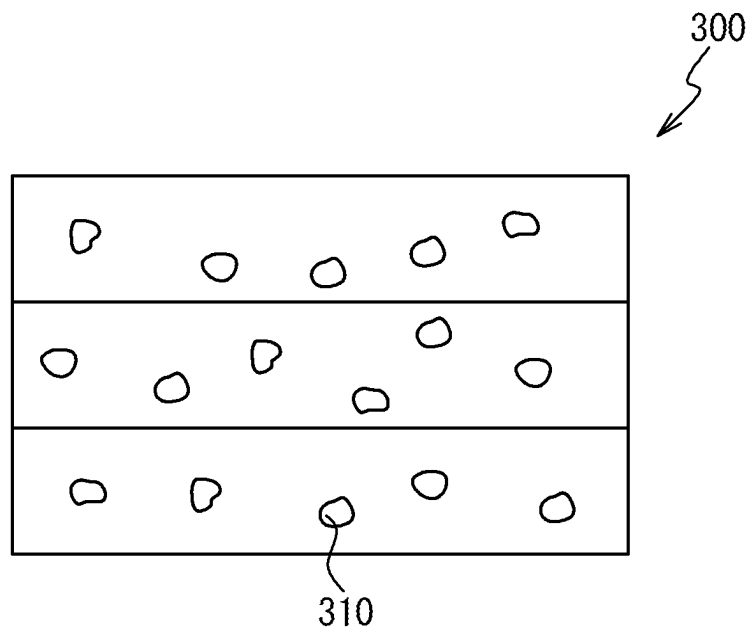
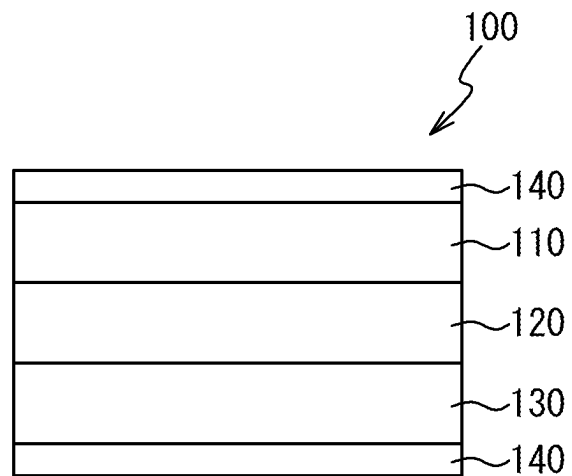


FIG. 11



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COMPOSITE MATERIAL, FLYING BODY AND COMPOSITE MATERIAL MANUFACTURING METHOD

TECHNICAL FIELD

The present invention relates to a composite material, a flying body and a composite material manufacturing method.

BACKGROUND

Patent Literature 1 discloses applying a composition to a component generated from a composite material in order to form a self-repairing layer on the component. The Patent Literature 1 further discloses that the composition includes a colloidal silica suspension, boron or boron compounds in powder form, a silicon carbide in powder form and at least a type of super heat resistant oxide.

CITED LIST

Patent Literature

[Patent Literature 1] Japanese patent application publication No. 2011-522099

SUMMARY

In view of the above situation, an objective is to provide a method of manufacturing a composite material with a plurality of characteristics, for example a high heat resistance and a high strength. Other objectives will be understood from following disclosures and descriptions of embodiments.

A composite material manufacturing method according to an embodiment to achieve the above objective includes: laminating a first sheet including a first slurry and a second sheet including a second slurry; and heating the first sheet and the second sheet that are laminated to a temperature of transforming to ceramics by pyrolysis to form an intermediate body. The manufacturing method further includes impregnating the intermediate body with a third slurry and heating at a temperature lower than the temperature of transforming to ceramics by pyrolysis. A composition of the first slurry is different from a composition of the second slurry.

A composite material according to an embodiment to achieve the above objective is provided with a first layer that includes a first matrix and a second layer that includes a second matrix and is provided below the first layer. The first layer includes a precursor of the first matrix. A composition of the second matrix is different from a composition of the first matrix.

A flying body according to an embodiment to achieve the above objective is provided with the above-mentioned composite material at a part where aerodynamic heating is applied.

According to the above embodiment, the composite material can have a plurality of characteristics.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of a composite material according to an embodiment.

FIG. 2A is a schematic cross-sectional view of a first layer according to an embodiment.

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FIG. 2B is a schematic diagram of A-A cross section in FIG. 2A.

FIG. 3 is a schematic cross-sectional view of a second layer according to an embodiment.

FIG. 4 is a schematic cross-sectional view of a third layer according to an embodiment.

FIG. 5 is a schematic diagram of a flying body according to an embodiment.

FIG. 6 is a flowchart that shows a composite material manufacturing method according to an embodiment.

FIG. 7 is a schematic diagram of a sheet used to manufacture a composite material in an embodiment.

FIG. 8 is a schematic diagram of a laminated body according to an embodiment.

FIG. 9 is a diagram to explain a pyrolysis of a laminated body in an embodiment.

FIG. 10 is a schematic diagram of an intermediate body according to an embodiment.

FIG. 11 is a schematic diagram of a composite material according to an embodiment.

DETAILED DESCRIPTION

(Embodiment) As shown in FIG. 1, a composite material **100** according to an embodiment is provided with a plurality of layers (including for example a first layer **110**, a second layer **120** and a third layer **130**). The composite material **100** is provided with an outer surface **100a** that is heated and an inner surface **100b** that faces the outer surface **100a**. When the outer surface **100a** of the composite material **100** is heated, an applied heat is transferred from the outer surface **100a** to the inner surface **100b** by heat conduction. The first layer **110** may form the outer surface **100a** of the composite material **100** and may be formed of materials of which a heat resistance is higher than other layers, for example the second layer **120** and/or the third layer **130**. The second layer **120** may be formed of a material of which a heat insulation performance is higher than other layers, for example the first layer **110** and/or the third layer **130**. The third layer **130** may form the inner surface **100b** of the composite material **100** and may be formed of a material which has a strength, for example a tensile strength, a bending strength at high temperature or the like, that is higher than other layers, for example the first layer **110** and/or the second layer **120**. The tensile strength is measured according to Japanese Industrial Standards (JIS) R 1606 for example. The bending strength at high temperature may be measured according to JIS R 1604 for example.

The first layer **110** includes reinforced fiber ceramics. For example, as shown in FIGS. 2A and 2B, the first layer **110** is provided with a first reinforced fiber **112**, a first matrix **114** and a first precursor **116**. The first reinforced fiber **112** includes for example a carbon fiber, a silicon carbide (SiC) fiber, a zirconium borohydride (ZrB₂) fiber or a combination thereof. The first matrix **114** is formed of ceramics in a porous shape. The first matrix **114** includes, for example, a material of Ultra High Temperature Ceramics (UHTC) that has a melting point or a decomposition temperature higher than or equal to 2750° C. (for example Zirconium boride (ZrB₂), hafnium carbide (HfC), hafnium boride (HfB₂)). In addition, the first matrix **114** may partially include a material that has a melting point or a decomposition temperature lower than 2750° C. such as SiC. The first precursor **116** is formed so as to impregnate into pores of the first matrix **114**. The first precursor **116** includes a substance before ceramics are formed. The first precursor **116** may include a substance that becomes ceramics when thermally decomposed, for

example Polycarbosilane (PCS). An UHTC may include ceramics of which a melting point or a decomposition temperature is higher than 2000° C. The decomposition temperature includes for example a temperature at which a substance is thermally decomposed.

The second layer **120** also includes reinforce fiber ceramics similarly to the first layer **110**. For example, as shown in FIG. 3, the second layer **120** is provided with a second reinforced fiber **122**, a second matrix **124** and a second precursor **126**. The second reinforced fiber **122** includes for example a carbon fiber, a SiC fiber, a ZrB₂ fiber or a combination thereof. The second matrix **124** is formed of ceramics in a porous shape. The second matrix **124** may include for example SiC or a material of UHTC that has a melting point or a decomposition temperature higher than or equal to 2750° C. (for example ZrB₂, HfC, Zirconium oxide (ZrO₂)). The second precursor **126** is formed so as to impregnate into pores of the second matrix **124**. The second precursor **126** includes a substance before ceramics are formed. The second precursor **126** may include a substance that becomes ceramics when thermally decomposed, for example PCS.

The third layer **130** also includes reinforced fiber ceramics similarly to the first layer **110**. For example, as shown in FIG. 4, the third layer **130** is provided with a third reinforced fiber **132**, a third matrix **134** and a third precursor **136**. The third reinforced fiber **132** includes for example a carbon fiber, a SiC fiber, a ZrB₂ fiber or a combination thereof. The third matrix **134** is formed of ceramics in a porous shape. The third matrix **134** may include for example SiC or a material of UHTC that has a melting point or a decomposition temperature higher than or equal to 2750° C. (for example ZrB₂, HfC). The third precursor **136** is formed so as to impregnate into pores of the third matrix **134**. The third precursor **136** includes a substance before ceramics are formed. The third precursor **136** may include a substance that becomes ceramics when thermally decomposed, for example PCS.

The reinforced fibers of each layer, for example the first reinforced fiber **112**, the second reinforced fiber **122** and the third reinforced fiber **132**, may be a same material or different materials. The precursors of each layer, for example the first precursor **116**, the second precursor **126** and the third precursor **136** may be a same material.

A heat resistance of the first layer **110** may be higher than a heat resistance of another layer, for example the second layer **120** and the third layer **130**. For example, a composition of the first matrix **114** (including for example materials and ratio of each material) may be different from a composition of a matrix of another layer, for example a composition of the second matrix **124** and/or a composition of the third matrix **134**. The material of the first matrix **114** may be different from the material of another matrix, and even if materials of the first matrix **114** is same as materials of a matrix of another layer, a ratio of each material, for example a mass percent concentration thereof, may be different. A ratio in the first layer **110** of a high heat resistance material, for example ZrB₂ and HfB₂, that are included in the first matrix **114**, for example a mass percent concentration thereof, may be higher than another layer, for example the second layer **120** and/or the third layer **130**. The high heat resistance material may include the UHTC. The high heat resistance material may include a material of which a melting point or a decomposition temperature is higher than a decomposition temperature of SiC, for example a UHTC of which a melting point or a decomposition temperature is higher than or equal to 2750° C. The high heat resistance

material may be a material of which a mass that decreases in unit time (for example reduction in thickness [g/sec]) when disposed in an environment where air is flowing at a high speed (for example a speed of sound) is lower than SiC, or may be a material of which a length of surface wear in unit time (for example a depletion rate [μm/sec]) is shorter than SiC. Herein, in addition, the length of surface wear in unit time indicates for example a length by which a thickness of a substance reduces during one second when the substance with a desired thickness is disposed in an environment where air is flowing at a high speed (for example a speed of sound). For example, the mass percent concentration of the high heat resistance material in the first layer **110** may be higher than or equal to 3%.

A heat resistance of the second layer **120** may be higher than a heat resistance of a layer in a direction from the second layer **120** to the inner surface **100b**, for example the third layer **130**. A composition of the second matrix **124** may be different from a composition of the third matrix **134**. A ratio, for example a mass percent concentration, in the second layer **120**, of a high heat resistance material, for example ZrO₂, that is included in the second matrix **124**, may be higher than a layer in a direction from the second layer **120** to the inner surface **100b**.

A thermal insulation performance of the second layer **120** may be higher than a thermal insulation performance of another layer, for example the first layer **110** and/or the third layer **130**. For example, a thermal conductivity of the second layer **120** is lower than a thermal conductivity of the first layer **110** and/or the third layer **130**. A thermal conductivity is measured for example according to JIS R1611, JIS R2251 or the like. A ratio, for example a mass percent concentration, of a high thermal insulation performance material, for example ZrO₂, that is included in the second matrix **124**, to the second layer **120**, may be higher than another layer, for example the first layer **110** and/or the third layer **130**. The high thermal insulation performance material may include a material of which a thermal conductivity is lower than SiC. The high thermal insulation performance material may include a material of which a thermal conductivity at a desired temperature, for example a room temperature or a high temperature (for example 2000° C.), is lower than SiC.

The composite material **100** can have a high heat resistance and a high strength by including a plurality of layers with different characteristics. The first layer **110** that forms the outer surface **100a** to be heated has a high heat resistance. The intermediate layer, for example the second layer **120**, has a high thermal insulation performance. The third layer **130** that forms the inner surface **100b** has a high strength. Thus, as the temperature of the inner surface **100b** is lower than the outer surface **100a** even if the outer surface **100a** is heated at a high temperature, a high strength material can be used to the third layer **130**.

The composite material **100** may be applied to a part of which one surface is heated and another surface is not heated. For example, as shown in FIG. 5, the composite material **100** may be provided to a tip part of a flying body **10** (for example a missile). The composite material **100** may be provided to a part where an aerodynamic heating is applied when the flying body **10** flies. The flying body **10** may be provided with a propulsion device **15**, for example a rocket motor or a jet engine, and fly at a speed faster than or equal to a speed of sound. The composite material **100** is provided so that an outer surface of the flying body **10** includes the outer surface **100a** of the composite material **100**.

The composite material **100** can have a high heat resistance and a high strength. As a heat resistance of the first layer **110** that forms the outer surface **100a** of the composite material **100** is high, the composite material **100** can resist to a high temperature due to an aerodynamic heating. In addition, as the third layer **130** of the composite material **100** has a high strength and is supporting the first layer **110**, a deformation of the first layer **110** due to an air pressure to the outer surface **100a** can be suppressed. As described above, the outer surface **100a** can keep an air flow even in an air flow, for example an air flow faster than or equal to a speed of sound. As a result, the composite material **100** can regulate an air flow without significantly affecting aerodynamics even in an air flow faster than or equal to a speed of sound.

(Manufacturing method) The composite material **100** can be manufactured by a method shown in FIG. 6. In step S110, a plurality of sheets in which a fiber cloth is impregnated with a slurry are prepared. For example, a first sheet **210** corresponding to the first layer **110**, a second sheet **220** corresponding to the second layer **120** and a third sheet **230** corresponding to the third layer **130** are prepared. A plurality of each of the first sheet **210**, the second sheet **220** and the third sheet **230** may be prepared.

For example, as shown in FIG. 7, the first sheet **210** corresponding to the first layer **110** is prepared. A first fiber cloth **212** formed of the first reinforced fiber **112** and a first slurry **214** to form the first matrix **114** are prepared. The first fiber cloth **212** is impregnated with the first slurry **214** to form the first sheet **210**. The first slurry **214** includes a precursor mixed with ceramic particles (for example SiC particles, UHTC particles of which a melting point or a decomposition temperature is higher than or equal to 2750° C.), for example a ceramic precursor. This ceramic precursor may include PCS. A plurality of first sheets **210** may be prepared.

For example, the second sheet **220** corresponding to the second layer **120** is prepared. A second fiber cloth **222** formed of the second reinforced fiber **122** and a second slurry **224** to form the second matrix **124** are prepared. The second fiber cloth **222** is impregnated with the second slurry **224** to form the second sheet **220**. The second slurry **224** includes a precursor mixed with ceramic particles (for example SiC particles, UHTC particles of which a melting point or a decomposition temperature is higher than or equal to 2750° C.), for example a ceramic precursor. This ceramic precursor may include PCS. A plurality of second sheets **220** may be prepared.

For example, the third sheet **230** corresponding to the third layer **130** is prepared. A third fiber cloth **232** formed of the third reinforced fiber **132** and a third slurry **234** to form the third matrix **134** are prepared. The third fiber cloth **232** is impregnated with the third slurry **234** to form the third sheet **230**. The third slurry **234** includes a precursor mixed with ceramic particles (for example SiC particles, UHTC particles of which a melting point or a decomposition temperature is higher than or equal to 2750° C.), for example a ceramic precursor. This ceramic precursor may include PCS. A plurality of third sheets **230** may be prepared.

In accordance with a change of ratio of ceramic particles included in the slurry, characteristics of each layer, for example a heat resistance, a thermal insulation performance or a strength, changes. A composition of the first slurry **214** may be different from a composition of another layer, for example a composition of the second slurry **224** and/or a composition of the third slurry **234**. A material of the first slurry **214** may be different from a material of a slurry of

another layer, and even if materials of the first slurry **214** are same as materials of a slurry of another layer, a ratio of each material, for example a mass percent concentration, may be different.

A heat resistance of a first sintered body in which the first slurry **214** is thermally decomposed may be higher than a sintered body in which another slurry is thermally decomposed, for example a second sintered body of the second slurry **224** and/or a third sintered body of the third slurry **234**. For example, a ratio, for example a mass percent concentration, of particles with high heat resistance in the first slurry **214**, for example ZrB₂ particles or HfB₂ particles, to the whole may be higher than another slurry, for example the second slurry **224** or the third slurry **234**.

A thermal insulation performance of the second sintered body in which the second slurry **224** is thermally decomposed may be higher than a sintered body in which another slurry is thermally decomposed, for example the first sintered body of the first slurry **214** and/or the third sintered body of the third slurry **234**. A ratio, for example a mass percent concentration, of particles with high thermal insulation performance in the second slurry **224**, for example ZrO₂ particles, to the whole may be higher than another slurry, for example the first slurry **214** and/or the third slurry **234**.

A strength of the third sintered body in which the third slurry **234** is thermally decomposed may be higher than a sintered body in which another slurry is thermally decomposed, for example the first sintered body of the first slurry **214** and/or the second sintered body of the second slurry **224**. A ratio, for example a mass percent concentration, of particles with high strength in the third slurry **234**, for example SiC particles, to the whole may be higher than another slurry, for example the first slurry **214** and/or the second slurry **224**.

In step S120, the sheets that are prepared are laminated and a laminated body **200** is formed. For example, as shown in FIG. 8, the first sheet **210** may be arranged on the second sheet **220** and the second sheet **220** may be arranged on the third sheet **230**. In addition, the second sheet **220** is arranged between the first sheet **210** and the third sheet **230**. A plurality of first sheets **210** may form the first layer **110**, a plurality of second sheets **220** may form the second layer **120** and a plurality of third sheets **230** may form the third layer **130**. The plurality of second sheets **220** may be arranged between the plurality of first sheets **210** and the plurality of third sheet **230**.

In step S130, the laminated body **200** is thermally decomposed by being heated in a vacuum atmosphere or an inert atmosphere (including for example a nitrogen atmosphere and an argon atmosphere) and an intermediate body **300** is formed. As shown in FIG. 9, the laminated sheets are heated in a state of being sandwiched between a forming mold **20** and a forming mold **30**. The forming mold **20** and the forming mold **30** may apply pressure to the laminated sheets. The sheets are heated to a temperature at which the first slurry **214**, the second slurry **224** and the third slurry **234** transform to ceramics by pyrolysis, for example a temperature between 800° C. and 1000° C. The forming mold **20** and the forming mold **30** include for example a metal mold, a carbon mold or the like.

In step S140, the intermediate body **300** that is thermally decomposed is impregnated with a slurry. As shown in FIG. 10, a plurality of pores **310** are formed in the intermediate body **300** in accordance with a ceramic yield. The intermediate body **300** is impregnated with slurry so as to fill the pores **310** of the intermediate body **300**. The intermediate

body **300** may be impregnated in a vacuum chamber adjusted to a vacuum atmosphere. The slurry with which the intermediate body **300** is impregnated may be any one of the first slurry **214**, the second slurry **224** and the third slurry **234**. The slurry with which the intermediate body **300** is impregnated may be a slurry different from the first slurry **214**, the second slurry **224** and the third slurry **234**. It should be noted that the ceramic yield may include a ratio between a mass of the ceramic precursor and a mass of the ceramics obtained by heating this ceramic precursor.

In step **S150**, a number of times of heating is compared to a desired number of times. When the number of times of heating has not reached the desired number of times (step **S150**: NO), the process returns to step **S130**. When the number of times of heating has reached a desired number of times (step **S150**: YES), the process shifts to step **S160**. The desired number of times is determined in accordance with characteristics of the composite material **100** to be manufactured. For example, the desired number of times may include a number less than or equal to 3 and may include 1. It should be noted that in a second and subsequent process of step **S130**, the forming molds **20** and **30** may not be used.

In step **S160**, the intermediate body **300** that is impregnated with the slurry is heated at a low temperature. The temperature to heat the intermediate body **300** is lower than the temperature at which the slurry transforms to ceramics by pyrolysis. The temperature of heating in step **S160** may be lower than a temperature at which a mass loss due to pyrolysis behavior becomes a desired ratio to a mass before heating. This desired ratio includes for example a ratio less than or equal to 10%. The desired ratio may be a ratio less than or equal to 5%. The ratio of the mass loss is measured by use of Thermal Gravimetric (TG) analysis, Thermal Gravimetric-Differential Thermal Analysis (TG-DTA) or the like. For example, when the immersed slurry is PCS, the temperature to heat may be less than 400° C.

As described above, by finally heating the intermediate body **300** that is impregnated with the slurry at a low temperature, a composite material **100** in which a composition of the first layer **110**, a composition of the second layer **120** and a composition of the third layer **130** are different can be manufactured. In addition, by keeping the temperature to heat low, a ratio of the pores **310** that occupy the composite material **100** can be suppressed.

By limiting the number of times of heating processes, the composite material **100** can include a plurality of layers with different characteristics. In step **140**, a same slurry is immersed in pores **310** formed in each layer of the intermediate body **300**. As a result, matrixes with a same composition are sintered in the pores **310** in each layer. By repeating heating processes and impregnations with slurry, compositions of matrixes in each layer become similar. By limiting the number of times of heating processes, each layer can have different characteristics.

Furthermore, by limiting the number of times of heating processes, a dense structure can be manufactured in few steps. As a result, a manufacturing cost is suppressed.

By including precursors (for example, the first precursor **116**, the second precursor **126**, the third precursor **136** and the like), the composite material **100** can have a high heat resistance. When the composite material **100** is heated, the first precursor **116**, the second precursor **126** and the third precursor **136** can be thermally decomposed and can absorb heat. In addition, by vaporization of the first precursor **116**, the second precursor **126** and the third precursor **136**, a heat input into the composite material **100** due to aerodynamic heating can be suppressed.

(Variation example) The composite material **100** may be composed of two layers, three layers or more. For example, the composite material **100** may be composed of the first layer **110** having a high heat resistance and the third layer **130** having a high strength. The composite material **100** may be provided with a layer having a heat resistance lower than the first layer **110** and higher than the second layer **120** between the first layer **110** and the second layer **120**.

As shown in FIG. **11**, the composite material **100** may be provided with an anti-hygroscopic coating **140**. The anti-hygroscopic coating **140** may include a coating that enhances water repellency, for example a polymer coating. The anti-hygroscopic coating **140** can suppress a deterioration, for example oxidation, decomposition or the like, of a precursor, for example the first precursor **116**, the second precursor **126** and/or the third precursor **136**, that the composite material **100** includes.

The embodiments and variation examples that are described above are examples and may be modified as long as the function is not interfered. In addition, the configurations described in each embodiment and each variation example can be arbitrarily modified and/or arbitrarily combined as long as the function is not interfered.

The composite material manufacturing method according to each embodiment is understood for example as follows.

A composite material manufacturing method according to a first aspect includes: laminating a first sheet (**210**) including a first slurry (**214**) and a second sheet (**230**) including a second slurry (**234**); and heating the first sheet (**210**) and the second sheet (**230**) that are laminated to a temperature of transforming to ceramics by pyrolysis to form an intermediate body (**300**). The manufacturing method further includes impregnating the intermediate body (**300**) with a third slurry and heating at a temperature lower than the temperature of transforming to ceramics by pyrolysis.

By heating the intermediate body (**300**) at a temperature lower than a temperature of transforming to ceramics by pyrolysis, the composite material (**100**) can be provided with a plurality of layers that have a plurality of characteristics. As a result, the composite material can have a plurality of characteristics, for example a high heat resistance and a high strength.

A composite material manufacturing method according to a second aspect is the composite material manufacturing method according to the first aspect and the lower temperature when heating the intermediate body (**300**) is configured to be a temperature at which a mass loss of the third slurry due to heating becomes in a desired ratio to a mass before heating.

A composite material manufacturing method according to a third aspect is the composite material manufacturing method according to the second aspect and the desired ratio of the mass loss of the third slurry due to heating to the mass before heating is configured to be a ratio less than or equal to 10%.

As the temperature when heating the intermediate body (**300**) is up to a temperature at which the mass loss of the third slurry becomes a desired ratio, the manufactured composite material (**100**) can include precursors (**116**, **126**, **136**). As a result, a ratio of pores (**310**) that occupy the composite material (**100**) can be suppressed.

A composite material manufacturing method according to a fourth aspect is the composite material manufacturing method according to the first aspect and is configured so that a heat resistance of a first sintered body in which the first

slurry (214) is thermally decomposed becomes higher than a second sintered body in which the second slurry (234) is thermally decomposed.

As a result, as the first layer (110) in which the first sheet (210) including the first slurry (214) is thermally decomposed has a high heat resistance, the composite material (100) has a high heat resistance to a heat applied from a surface (100a) of the first layer (110).

A composite material manufacturing method according to a fifth aspect is the composite material manufacturing method according to the fourth aspect and is configured so that a strength of the second sintered body becomes higher than a strength of the first sintered body.

As a result, as the second layer (130) in which the second sheet (230) including the second slurry (234) is thermally decomposed has a high strength, the second layer (130) supports the first layer (110) and the composite material (100) has a high strength.

A composite material manufacturing method according to a sixth aspect is the composite material manufacturing method according to the first aspect and the laminating includes laminating a third sheet (220) including a fourth slurry (224) between the first sheet (210) and the second sheet (230). Heating the first sheet (210) and the second sheet (220) includes heating the third sheet (220) to a temperature of transforming to ceramics. In addition, a composition of the fourth slurry (224) is different from a composition of the first slurry (214) and a composition of the second slurry (234).

As a result, the manufactured composite material can include three or more layers having different characteristics.

A composite material manufacturing method according to a seventh aspect is the composite material manufacturing method according to the sixth aspect and is configured so that a thermal insulation performance of a third sintered body in which the fourth slurry (224) is thermally decomposed becomes higher than a thermal insulation performance of the first sintered body in which the first slurry (214) is thermally decomposed.

A composite material manufacturing method according to an eighth aspect is the composite material manufacturing method according to the sixth aspect and is configured so that the thermal insulation performance of the third sintered body in which the fourth slurry (224) is thermally decomposed is higher than a thermal insulation performance of the second sintered body in which the second slurry (234) is thermally decomposed.

As the manufactured composite material includes the third layer (120) with a high thermal insulation performance between the first layer (110) and the second layer (130), a conduction of heat of the first layer (110) that is heated to the second layer (130) can be suppressed.

The composite material according to each embodiment is understood for example as follows.

A composite material according to a ninth aspect is provided with a first layer (110) that includes a first matrix (114) and a second layer (130) that includes a second matrix (134) and is provided under the first layer (110). The first layer (110) includes a precursor of the first matrix (114). A composition of the second matrix (134) is different from a composition of the first matrix (114).

The composite material can be provided with a plurality of layers that have a plurality of characteristics. As a result, the composite material can have a plurality of characteristics, for example a high heat resistance and a high strength. In addition, as the first layer (110) includes a precursor of the

first matrix (114), a ratio of pores (310) that occupy the composite material (100) can be suppressed.

The composite material according to a tenth aspect is the composite material according to the ninth aspect and is configured so that a heat resistance of the first layer (110) is higher than a heat resistance of the second layer (130).

The composite material has a high heat resistance to a heat applied from a surface (100a) of the first layer (110).

A composite material according to an eleventh aspect is the composite material according to the ninth aspect and is configured so that a strength of the second layer (130) is higher than a strength of the first layer (110).

As the second layer (130) has a high strength, the second layer (130) supports the first layer (110) and the composite material has a high strength.

A composite material according to a twelfth aspect is the composite material according to the ninth aspect and is further provided with a third layer (120) that includes a third matrix (134) and is provided between the first layer (110) and the second layer (130). A thermal insulation performance of the third layer (120) is higher than a thermal insulation performance of the first layer (110).

A composite material according to a thirteenth aspect is the composite material according to the twelfth aspect and is configured so that the thermal insulation performance of the third layer (120) becomes higher than a thermal insulation performance of the second layer (130).

As the composite material includes the third layer (120) with a high thermal insulation performance between the first layer (110) and the second layer (130), a conduction of heat of the first layer (110) that is heated to the second layer (130) can be suppressed.

A composite material according to a fourteenth aspect is the composite material according to the ninth aspect and is configured so that the first layer (110) forms an outer surface (100a) and the outer surface (100a) regulates an air flow.

A composite material according to a fifteenth aspect is the composite material according to the fourteenth aspect and is configured so that the outer surface (100a) regulates the air flow that is faster or equal to a speed of sound.

The flying body according to each embodiment is understood for example as follows.

A flying body according to a sixteenth aspect is provided with the composite material according to the ninth aspect at a part where an aerodynamic heating is applied.

It should be noted that the present application claims priority based on Japanese Patent Application No. 2020-033742 filed on Feb. 28, 2020, the whole disclosure of which is incorporated herein by reference.

The invention claimed is:

1. A composite material manufacturing method including: laminating a first sheet including a first slurry and a second sheet including a second slurry;

heating the first sheet and the second sheet that are laminated to a first temperature at which the first slurry and the second slurry transform to ceramics by pyrolysis to form an intermediate body; and

impregnating the intermediate body with a third slurry and heating the impregnated intermediate body at a second temperature, the second temperature being lower than a third temperature at which the third slurry transforms to ceramics by pyrolysis,

wherein a first composition of the first slurry is different from a second composition of the second slurry, and wherein the first slurry includes a ceramic precursor, the second slurry includes a ceramic precursor, and the third slurry includes a ceramic precursor.

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2. The composite material manufacturing method according to claim 1,

wherein the second temperature is a temperature at which a mass loss of the third slurry due to heating becomes in a desired ratio to a mass of the third slurry before heating.

3. The composite material manufacturing method according to claim 2,

wherein the desired ratio is a ratio less than or equal to 10%.

4. The composite material manufacturing method according to claim 1, wherein a first heat resistance of a first sintered body in which the first slurry is thermally decomposed is higher than a second heat resistance of a second sintered body in which the second slurry is thermally decomposed.

5. The composite material manufacturing method according to claim 4,

wherein a second strength of the second sintered body is higher than a first strength of the first sintered body.

6. The composite material manufacturing method according to claim 1,

wherein the laminating includes laminating a third sheet including a fourth slurry between the first sheet and the second sheet,

wherein the heating of the first sheet and the second sheet includes heating the third sheet to the first temperature, the first temperature being a temperature at which the fourth slurry transforms to ceramics by pyrolysis,

wherein a third composition of the fourth slurry is different from the first composition of the first slurry and the second composition of the second slurry, and

wherein the fourth slurry includes a ceramic precursor.

7. The composite material manufacturing method according to claim 6, wherein a second thermal insulation performance of a second sintered body in which the fourth slurry is thermally decomposed is higher than a first thermal insulation performance of a first sintered body in which the first slurry is thermally decomposed.

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8. The composite material manufacturing method according to claim 6,

wherein a first thermal insulation performance of a first sintered body in which the fourth slurry is thermally decomposed is higher than a second thermal insulation performance of a second sintered body in which the second slurry is thermally decomposed.

9. A flying body comprising a composite material at a part where aerodynamic heating is applied, the composite material comprising:

a first layer that includes a first matrix; and

a second layer that includes a second matrix and is provided below the first layer,

wherein the first layer includes a precursor of the first matrix, and

wherein the first matrix includes Ultra High Temperature Ceramics (UHTC) particles that have a melting point or a decomposition temperature greater than or equal to 2750° C., a composition of the second matrix is different from a composition of the first matrix, and the first matrix includes more UHTC particles than the second matrix.

10. The flying body according to claim 9,

wherein a heat resistance of the first layer is higher than a heat resistance of the second layer.

11. The flying body according to claim 9,

wherein a strength of the second layer is higher than a strength of the first layer.

12. The flying body according to claim 9, further comprising a third layer with a third matrix, the third layer provided between the first layer and the second layer,

wherein a heat resistance of the third layer is higher than a heat resistance of the first layer.

13. The flying body according to claim 12,

wherein the heat resistance of the third layer is higher than a heat resistance of the second layer.

14. The flying body according to claim 9,

wherein the first layer forms an outer surface, and wherein the outer surface regulates an air flow.

15. The flying body according to claim 14,

wherein the air flow is faster than or equal to a speed of sound.

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